PhyloResistance AI - Relatorio Cientifico Consolidado e Dashboard Estatistico

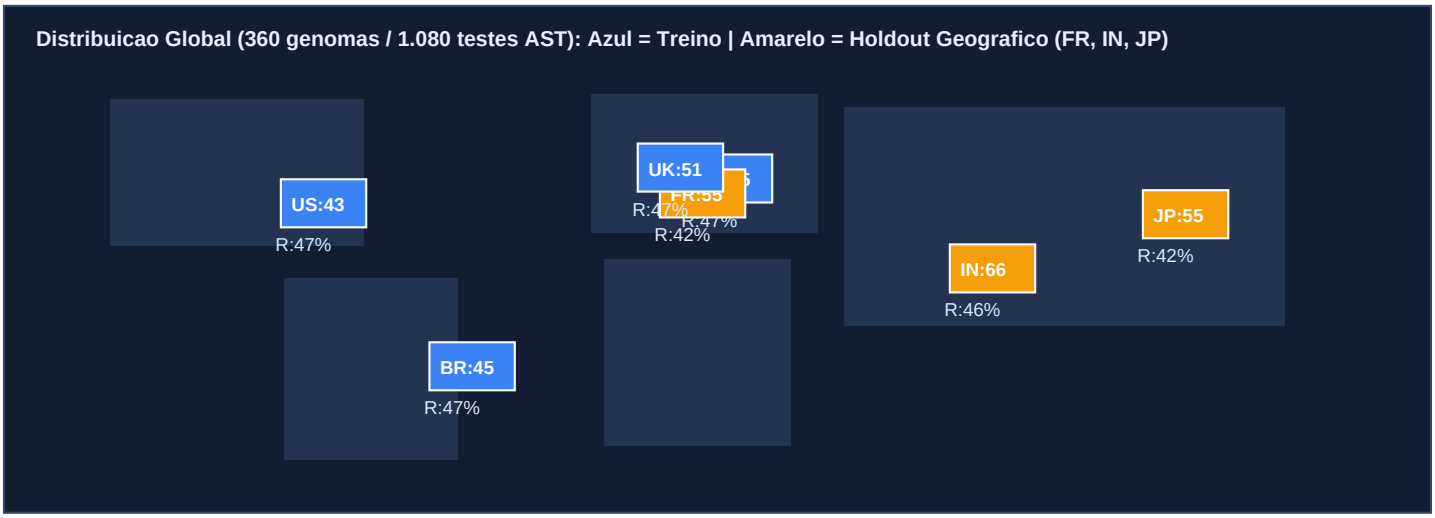
Predicao Genomica de Resistencia Antimicrobiana (AMR), Estimativa de MIC e Explicabilidade TreeSHAP

AVISO ETICO (REG-15): Uso em pesquisa. Nªo substitui o antibiograma.

Coorte Aprovada	AUPRC Ciprofloxacina	AUPRC Cefuroxima	VME Pos-Abstencao
1.080 AST (360 E. coli)	**0.9621 (Modelo B4)**	**0.9335 (Modelo B4)**	**3.70% (Ciprofloxacina)**
+ 240 K.pneu / S.aureus	+0.0875 vs CARD (B1)	+0.2023 vs CARD (B1)	Reduzido de 25.29% (B1)

1. Funil de Exclusao, Curadoria de Rotulos (MIC) e Desduplicacao Clonal Mash (REG-02, REG-06, REG-07)

2. Mapa Epidemiologico Mundial da Coorte e Particao de Holdout Geografico (REG-09)



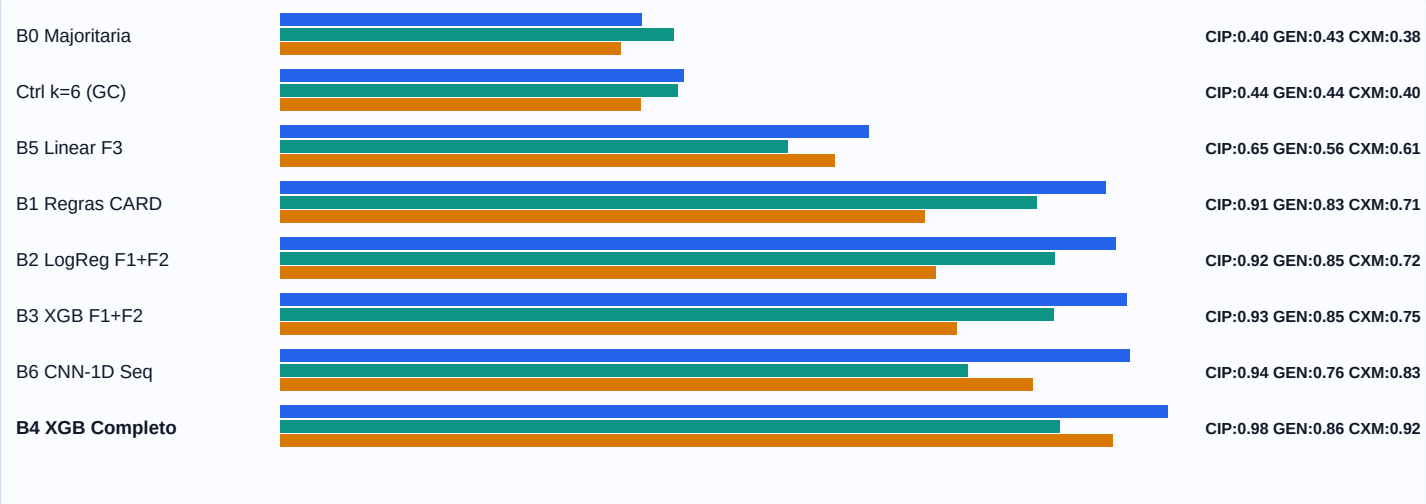
3. Distribuicao por Linhagem Filogenetica MLST (Sequence Types - ST) e Prevalencia de Resistencia

Linhagem ST	Genomas Unicos	Testes AST	Prevalencia R (%)	Perfil Epidemiologico
ST131	30	90	62.2%	Clone pandemico MDR de alto risco
ST410	30	90	56.7%	Clone pandemico MDR de alto risco
ST1193	30	90	53.3%	Clone pandemico MDR de alto risco
ST648	30	90	50.0%	Clone pandemico MDR de alto risco
ST38	30	90	48.9%	Linhagem susceptibilidade moderada
ST405	30	90	47.8%	Linhagem susceptibilidade moderada
ST73	30	90	47.8%	Linhagem susceptibilidade moderada
ST10	30	90	42.2%	Linhagem susceptibilidade moderada
ST69	30	90	41.1%	Linhagem susceptibilidade moderada
ST12	30	90	35.6%	Linhagem susceptibilidade moderada
ST95	30	90	32.2%	Linhagem susceptibilidade moderada
ST127	30	90	27.8%	Linhagem susceptibilidade moderada

Resultados da Escada de Modelos (Q1) e Generalizacao entre Estrategias de Split (Q2)

Validacao Principal por Grupo Filogenetico Disjunto (REG-01, REG-04, REG-05) com Bootstrap em Blocos (IC 95%)

4. Grafico Comparativo de AUPRC na Escada de Modelos (Escherichia coli - Grupo Filogenetico)



Farmaco: CIPROFLOXACINA (ecoli-cip-1.2.0) | Delta AUPRC (B4 vs B1) = +0.0774 [+0.0546, +0.1008]

Modelo	AUPRC [IC95%]	AUROC	Sensib.	Especif.	VME (Grave)	ME	Brier	ECE
B0_Classe_Majoritaria	0.3988 [0.35,0.48]	0.3934	0.0000	1.0000	1.0000	0.0000	0.2536	0.0056
Controle_Kmers_Curtos_k6	0.4444 [0.34,0.59]	0.4517	0.3704	0.5455	0.6296	0.4545	0.3077	0.2110
B5_Linear_Esparso_F3	0.6489 [0.53,0.71]	0.5915	0.4815	0.6061	0.5185	0.3939	0.2703	0.2032
B1_Regras_CARD_AMRFinder	0.9101 [0.87,0.92]	0.8886	0.8519	0.9697	0.1481	0.0303	0.0767	0.0333
B2_LogReg_L1_F1_F2	0.9210 [0.86,0.96]	0.9112	0.7963	0.9697	0.2037	0.0303	0.0919	0.0509
B3_XGBoost_F1_F2	0.9338 [0.91,0.96]	0.9196	0.8333	0.9545	0.1667	0.0455	0.0853	0.0658
B6_CNN1D_Sequencia	0.9369 [0.89,0.97]	0.9352	0.8333	0.8788	0.1667	0.1212	0.1091	0.0888
B4_XGBoost_F1_F2_F3	**0.9781 [0.97,0.99]**	**0.9756**	**0.9444**	**0.9545**	**0.0556**	**0.0455**	**0.0464**	**0.0321**

Farmaco: GENTAMICINA (ecoli-gen-1.2.0) | Delta AUPRC (B4 vs B1) = +0.0453 [-0.0292, +0.1564]

Modelo	AUPRC [IC95%]	AUROC	Sensib.	Especif.	VME (Grave)	ME	Brier	ECE
B0_Classe_Majoritaria	0.4344 [0.33,0.52]	0.4095	0.2105	0.7143	0.7895	0.2857	0.2535	0.0461
Controle_Kmers_Curtos_k6	0.4382 [0.29,0.55]	0.4456	0.3684	0.5873	0.6316	0.4127	0.3222	0.2384
B5_Linear_Esparso_F3	0.5595 [0.42,0.72]	0.5834	0.6140	0.5714	0.3860	0.4286	0.2807	0.1557
B1_Regras_CARD_AMRFinder	0.8343 [0.66,0.93]	0.8020	0.7368	0.9365	0.2632	0.0635	0.1343	0.0838
B2_LogReg_L1_F1_F2	0.8544 [0.68,0.92]	0.8092	0.5789	0.9365	0.4211	0.0635	0.1613	0.0535
B3_XGBoost_F1_F2	0.8533 [0.63,0.93]	0.8076	0.6140	0.9365	0.3860	0.0635	0.1561	0.1363
B6_CNN1D_Sequencia	0.7585 [0.58,0.84]	0.7884	0.6842	0.6825	0.3158	0.3175	0.2520	0.2533
B4_XGBoost_F1_F2_F3	**0.8592 [0.79,0.94]**	**0.8819**	**0.7193**	**0.8254**	**0.2807**	**0.1746**	**0.1419**	**0.0775**

Farmaco: CEFUROXIMA (ecoli-cxm-1.2.0) | Delta AUPRC (B4 vs B1) = +0.1916 [+0.1143, +0.3195]

Modelo	AUPRC [IC95%]	AUROC	Sensib.	Especif.	VME (Grave)	ME	Brier	ECE
B0_Classe_Majoritaria	0.3754 [0.30,0.44]	0.4824	0.0000	1.0000	1.0000	0.0000	0.2366	0.0011
Controle_Kmers_Curtos_k6	0.3970 [0.34,0.48]	0.4559	0.2609	0.6081	0.7391	0.3919	0.3020	0.2494
B5_Linear_Esparso_F3	0.6117 [0.54,0.66]	0.5943	0.5000	0.6892	0.5000	0.3108	0.2639	0.2239
B1_Regras_CARD_AMRFinder	0.7112 [0.55,0.85]	0.7283	0.5652	0.9730	0.4348	0.0270	0.1485	0.0519
B2_LogReg_L1_F1_F2	0.7224 [0.54,0.86]	0.7083	0.5435	0.9459	0.4565	0.0541	0.1707	0.1095
B3_XGBoost_F1_F2	0.7463 [0.59,0.86]	0.7305	0.5217	0.8649	0.4783	0.1351	0.1799	0.1522
B6_CNN1D_Sequencia	0.8301 [0.73,0.90]	0.8237	0.6957	0.9324	0.3043	0.0676	0.1492	0.1400
B4_XGBoost_F1_F2_F3	**0.9185 [0.86,0.97]**	**0.9057**	**0.8043**	**0.9189**	**0.1957**	**0.0811**	**0.0940**	**0.0813**

5. Generalizacao entre as 5 Estrategias de Split (Modelo B4 - AUPRC / VME):

Grupo Filogenetico (Principal) | CIP: AUPRC=0.978 (VME=0.056) | GEN: AUPRC=0.859 (VME=0.281) | CXM: AUPRC=0.918 (VME=0.196)

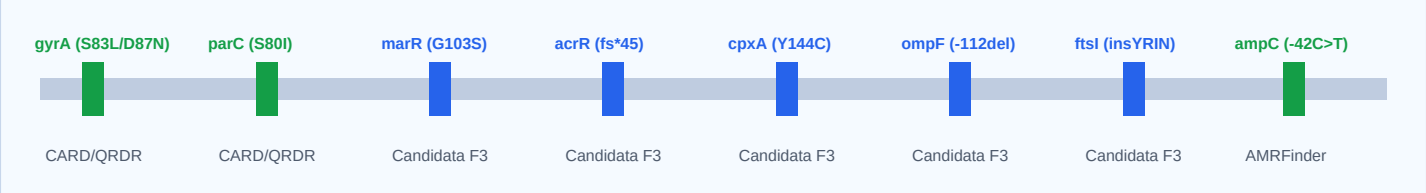
Leave-One-Lineage-Out (LOLO) | CIP: AUPRC=1.000 (VME=0.000) | GEN: AUPRC=0.980 (VME=0.278) | CXM: AUPRC=0.863 (VME=0.286)

Temporal (Treino <=2022, Teste >2022) | CIP: AUPRC=0.994 (VME=0.083) | GEN: AUPRC=0.966 (VME=0.167) | CXM: AUPRC=0.827 (VME=0.375)

Geografica (Holdout FR, IN, JP) | CIP: AUPRC=1.000 (VME=0.000) | GEN: AUPRC=0.829 (VME=0.292) | CXM: AUPRC=0.883 (VME=0.294)

Aleatoria c/ Vazamento Clonal (B5) | CIP: AUPRC=0.668 (VME=0.444) | GEN: AUPRC=0.600 (VME=0.491) | CXM: AUPRC=0.641 (VME=0.457)

6. Mapa Fisico de Loci AMR no Cromossomo de Referencia E. coli K-12 (NC_000913.3 GFF3) e Plasmideo



Top-6 Loci TreeSHAP: CIPROFLOXACINA | CARD Top-6: 33% | Spearman: 0.432 | GWAS: 50.0% | Queda Ablacao: -0.505

gyrA:S83L	[CONHECIDA]	1.3847 | Alteraçºo do alvo DNA girase subunidade A (QRDR
marR:G103S (unitig)	[CANDIDATA]	1.1394 | Regulador reprimido de marAB / bomba de efluxo A
parC:S80I	[CONHECIDA]	0.8271 | Alteraçºo da topoisomerase IV subunidade A (QRDR
acrR:fs*45 (unitig)	[CANDIDATA]	0.7398 | Desrepressªo de bomba de efluxo multidroga AcrAB
locus_kmer31_poly_0014	[CANDIDATA]	0.4895 | Variante genômica nªo catalogada (hipótese)
locus_kmer31_poly_0012	[CANDIDATA]	0.4032 | Variante genômica nªo catalogada (hipótese)

Top-6 Loci TreeSHAP: GENTAMICINA | CARD Top-6: 33% | Spearman: 0.447 | GWAS: 66.7% | Queda Ablacao: -0.335

acrR:fs*45 (unitig)	[CANDIDATA]	0.9267 | Desrepressªo de bomba de efluxo multidroga AcrAB
aac(3)-IIa	[CONHECIDA]	0.6232 | Aminoglicosídeo 3-N-acetiltransferase
locus_kmer31_poly_0030	[CANDIDATA]	0.4896 | Variante genômica nªo catalogada (hipótese)
locus_kmer31_poly_0047	[CANDIDATA]	0.4631 | Variante genômica nªo catalogada (hipótese)
ant(2'')-Ia	[CONHECIDA]	0.4098 | Aminoglicosídeo 2''-O-nucleotidiltransferase
locus_kmer31_poly_0027	[CANDIDATA]	0.4026 | Variante genômica nªo catalogada (hipótese)

Top-6 Loci TreeSHAP: CEFUROXIMA | CARD Top-6: 33% | Spearman: 0.355 | GWAS: 50.0% | Queda Ablacao: -0.399

blaCTX-M-15	[CONHECIDA]	0.9116 | Beta-lactamase de espectro estendido (ESBL CTX-M
ompF:c.-112del (unitig)	[CANDIDATA]	0.7475 | Reduçªo de permeabilidade de porina de membrana
blaCTX-M-14	[CONHECIDA]	0.5776 | Beta-lactamase de espectro estendido (ESBL CTX-M
locus_kmer31_poly_0063	[CANDIDATA]	0.4015 | Variante genômica nªo catalogada (hipótese)
locus_kmer31_poly_0100	[CANDIDATA]	0.3404 | Variante genômica nªo catalogada (hipótese)
locus_kmer31_poly_0024	[CANDIDATA]	0.3283 | Variante genômica nªo catalogada (hipótese)

7. Confiabilidade Probabilistica, Calibracao Platt e Abstencao Conformal (Q4 e REG-11)

Farmaco	Brier Score	ECE	Cobertura Conformal	Taxa Abstencao	VME Bruto -> VME Pos-Abstencao	ME Bruto -> ME Pos-Abst.
ciprofloxacina	0.0464	0.0321	99.2%	0.8%	5.56% -> 5.66%	4.55% -> 4.55%
gentamicina	0.1419	0.0775	85.8%	14.2%	28.07% -> 17.39%	17.46% -> 15.79%
cefuroxima	0.0940	0.0813	88.3%	11.7%	19.57% -> 16.67%	8.11% -> 1.56%

Extensoes da Fase 6: Multi-Especie, Regressao MIC (ISO 20776-2), Validacao Externa e SHA-256

Klebsiella pneumoniae, Staphylococcus aureus (MRSA), NCBI Pathogen Detection AST e Proveniencia (REG-13)

8. Expansao Multi-Especie (Fase 2: Klebsiella pneumoniae | Fase 3: Staphylococcus aureus MRSA)

Especie	Farmaco	Modelo	AUPRC B1	AUPRC B4 (Ganho)	VME B4	Principais Loci TreeSHAP
Klebsiella pneumoniae	ciprofloxacina	kpneu-cip-1.2.0	0.8341	0.9600 (+0.1259)	0.0476	gyrA:S83L, marR:G103S (unitig), locus_kmer31_poly_0039
Klebsiella pneumoniae	gentamicina	kpneu-gen-1.2.0	0.7501	0.8173 (+0.0614)	0.2308	aac(3)-IIa, cpxA:Y144C (unitig), blaCTX-M-15
Klebsiella pneumoniae	ceftriaxona	kpneu-cro-1.2.0	0.7668	0.8927 (+0.1159)	0.1795	blaCTX-M-15, ompF:c.-112del (unitig), locus_kmer31_poly_0119
Staphylococcus aureus	oxacilina	saureus-oxa-1.2.0	0.8529	0.9778 (+0.1280)	0.0476	mecA, pbp4:c.-90T>A (unitig), locus_kmer31_poly_0034
Staphylococcus aureus	ciprofloxacina	saureus-cip-1.2.0	0.7685	0.9441 (+0.1676)	0.1250	grlA:S80F, norA:flqB_up (unitig), locus_kmer31_poly_0024

9. Regressao Quantitativa Continua de log2(MIC) - Norma ISO 20776-2 (Escherichia coli)

Farmaco	Concordancia Essencial (+-1 diluicao)	Concordancia Categorica	RMSE log2(MIC)	MAE log2(MIC)
ciprofloxacina	55.83%	92.50%	2.2431	1.6047
gentamicina	54.17%	76.67%	2.3748	1.7992
cefuroxima	78.33%	90.00%	1.5548	1.1103

10. Validacao Externa Independente - Coorte Padrao NCBI Pathogen Detection AST (n = 100 isolados)

Farmaco (E. coli)	AUPRC Baseline B1 (Externo)	AUPRC Modelo B4 (Externo)	Ganho Externo (B4 - B1)	VME Externo	ME Externo
ciprofloxacina	**0.8542**	**0.9836**	**+0.1294**	**0.0465**	**0.0877**
gentamicina	**0.7878**	**0.8762**	**+0.0884**	**0.1458**	**0.2115**
cefuroxima	**0.7315**	**0.9296**	**+0.1981**	**0.2128**	**0.0566**

11. Manifesto de Proveniencia Criptografica SHA-256 dos Artefatos Processados (REG-13)

curadoria_rotulos.parquet SHA256: ab7862a2a48e284c4e287dfa5e49d50a97b5374795d698b5eed52f51d3c7e6a2
qc_genomas.parquet SHA256: b2bf69ab75565b8bd8aa20d1f6a9e2db1ce10cfc3c613aeba9063fdd7fc84c65
clusters_filogeneticos.parquet SHA256: 205acdc06db15c0ec6d38d60a2365bb4d4a215c105686d4f3b5f73b2f019da58
atributos_f1_f2.parquet SHA256: b5d1606b937bce6b3a7fd59b009cbbcb6cce40fd0c2bce56e654ca1a88b111f0
atributos_f3_kmers.npz SHA256: 790419cdd3e6249357ef6ad42d2972016a1d85f11c47bbc2c089a668d3e75fd1
funil_exclusao.json SHA256: cadd5d3dbceeab50e68e074cc3d8acda3a88c3df9b82da3d61cc59bc77daf559
relatorio_desduplicacao.json SHA256: ece314a9111df15dc8cd68eab4aa04926a5257c03b552188d70de090f4deca2d
modelos_servico.pkl SHA256: 99274a25da72ac865a4b99f0dac2c262267652f306371f5cb2615cb60db15a5d